A SEPARATED MORPHISM OF ALGEBRAIC SPACES SATISFIES THE UNIQUENESS PART OF THE VALUATIVE CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a clean equalizer argument using the domain property of valuation rings; it is short but every step is present.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The shape of the valuative criterion diagram and its uniqueness part. Definition definition-valuative-criterion, used at proof lines 9192-9195. Fixes the situation in which the two morphisms a and b are considered.
- Separatedness of f means the diagonal is a closed immersion. used at proof lines 9198-9199. Makes the equalizer Z a closed subscheme of Spec(A).
- The equalizer of two morphisms as a fibre product along the diagonal. used at proof lines 9196-9198. Identifies Z and its underlying closed set.
- A valuation ring is a domain, so a closed subscheme of Spec(A) containing the generic point is everything. used at proof lines 9199-9201. Forces Z = Spec(A) and hence a = b.

Source reference. Original definition 03IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4619–4619. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is separated, then f satisfies the uniqueness part of the valuative criterion.*

Proof. Let a diagram as in Definition 03IX be given. Suppose there are two distinct morphisms $a, b : \mathrm{Spec}(A) \to X$ fitting into the diagram. Let $Z \subset \mathrm{Spec}(A)$ be the equalizer of a and b. Then $Z = \mathrm{Spec}(A) \times_{(a,b),X\times_Y X,\Delta} X$. If f is separated, then Δ is a closed immersion, and this is a closed subscheme of $\mathrm{Spec}(A)$. By assumption it contains the generic point of $\mathrm{Spec}(A)$. Since A is a domain this implies $Z = \mathrm{Spec}(A)$. Hence $a = b$ as desired. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.